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PoseD-Flow: Versatile and Guided Flow Matching Model of Human Pose

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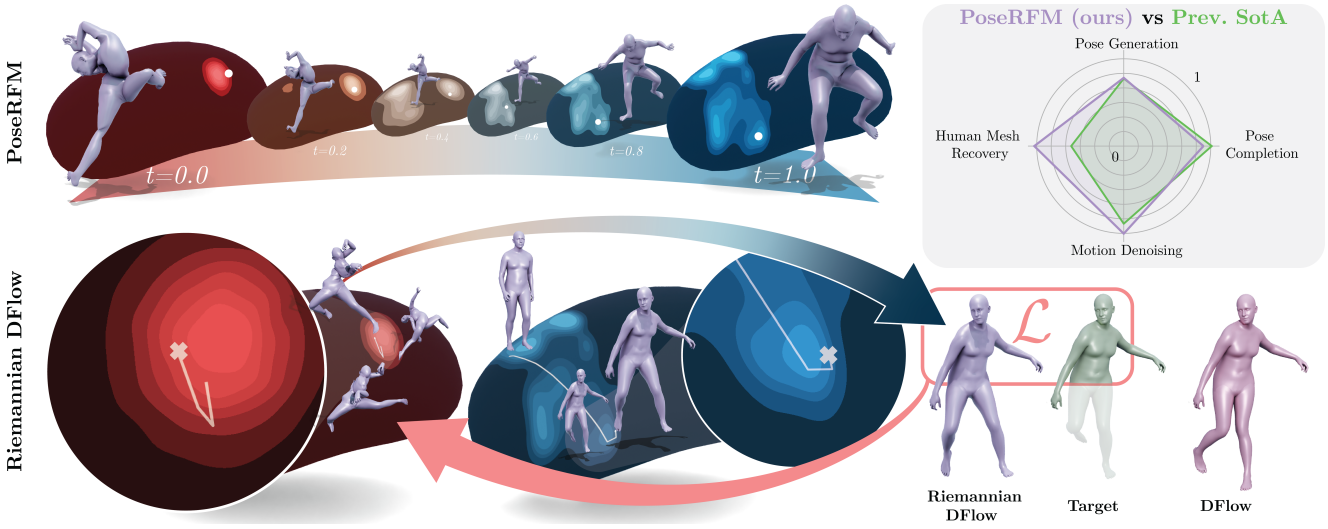


Figure 1. **PoseD-Flow** framework: (**top**) **PoseRFM**, a robust human pose prior defined on the product manifold of joints using Riemannian Flow Matching; (**bottom**): **Riemannian D-Flow**, a flexible, geometry-aware inversion technique for flow models. Together, they provide a novel approach to solving inverse problems in human pose, achieving results competitive with SotA diffusion models.

Abstract

Generative pose priors have recently emerged as a powerful tool for inference under occlusion or noise. Yet today's strongest generative paradigm, flow matching*, remains unused for human pose due to two fundamental barriers: the absence of a pre-trained flow prior and the non-Euclidean nature of articulated poses. We overcome both by introducing PoseD-Flow, a novel framework to unify Riemannian Flow Matching (RFM) with training-free guidance for 3D human pose recovery. PoseD-Flow is composed of two contributions: (i) PoseRFM, the first RFM model of human pose, defined directly on the product manifold of joint rotations, and (ii) Riemannian D-Flow, a principled guidance mechanism that, by differentiating through its ODE sampling dynamics, conditions PoseRFM at inference without any task-specific training. Our theoretical analysis shows that the induced dynamics are shaped by data covariance and manifold curvature, yielding a bias toward realistic poses. Across pose completion, denoising, and inverse kinematics, PoseD-Flow establishes new state of the art, particularly under noise, occlusion, and partial observations.*

1. Introduction

Humans do not merely occupy environments. We co-create them. Our physical configuration is our primary interface with the world, mediating perception, intent, and interaction. This configuration, **human pose**, is the instantaneous, deliberate spatial arrangement of the body in 3D. For machines to operate in human-centric environments, reasoning about pose is not optional but essential. Yet, faithful 3D pose estimation remains profoundly challenging: visual observations are noisy, partial, ambiguous, and frequently occluded, demanding *priors* that understand *what valid human pose actually looks like*.

The rapid evolution of deep generative modeling has opened an enticing direction: learning expressive pose priors that can later be used to infer plausible 3D configurations from incomplete 2D or 3D evidence [16, 26, 46, 54, 60]. Among generative paradigms, diffusion [27, 58] and flow matching [40, 41] have emerged as state of the art for modeling complex distributions. However, using these models as inference engines, *e.g.*, recovering poses that best explain given observations, requires *inverting* a generative process not trained for conditioning. This demands *training-free guidance*, a notoriously delicate prob-

lem [18]. While recent efforts have begun to explore guidance [4, 21, 68, 73], principled inversion of unconditional generative models remains unresolved for human pose.

Several approaches attempt pose recovery by optimizing in latent or geometry spaces at test time: HuMoR [54] inverts conditional VAEs, D-Poser [46] optimizes through diffusion, while PoseNDF [60] and NRDF [26] iteratively project onto neural distance fields [15]. Despite their success, none leverage the most expressive and tractable generative paradigm today, *flow matching* [40], nor do they treat human pose in its true configuration space: *a non-Euclidean product manifold of 3D rotations*.

We bridge this gap by proposing **PoseD-Flow**, the first framework to unite geometric flow-based pose priors with training-free guidance. As illustrated in Fig. 1, our method consists of two key components: (**i**) **PoseRFM**: the first Riemannian Flow Matching model of human pose, which models pose directly on the appropriate manifold of articulated rotations rather than in unconstrained Euclidean space; (**ii**) **Riemannian D-Flow**: a novel, training-free, geometry-respecting guidance mechanism that enables conditional generation by back-propagating through the Riemannian-ODE sampling process, yielding a theoretically grounded and empirically powerful strategy for solving pose recovery as an inverse problem. Unlike classifier-free or heuristic guidance methods, Riemannian D-Flow exposes a deeper inductive mechanism: gradients propagate through the flow in ways governed by data covariance and manifold curvature, creating a natural bias toward realistic, stable pose solutions. The result is a model that not only generates plausible poses, but *inverts* reliably, even under severe occlusions, noise, and partial observations.

Across denoising, pose completion, and inverse kinematics, PoseD-Flow sets a new state of the art, improving both geometric accuracy and perceptual plausibility, while being fully training-free at inference time. Our contributions are:

- **PoseRFM**, the first large-scale, Riemannian Flow Matching (RFM) model of human pose, supporting training-free, geometry-aware inversion on the articulation space;
- **Riemannian D-Flow**, a principled, training-free source-point optimization framework that guides any RFM model via differentiation through Riemannian ODE sampling, preserving geometry and stability;
- **Theoretical insights**, revealing that inference dynamics enjoy an inductive bias shaped by both data covariance and manifold curvature, offering an explanation for robustness against noise and ambiguity;
- **PoseD-Flow**, a versatile framework for human pose estimation, surpassing prior methods on motion denoising, completion, and inverse kinematics, particularly under occlusion and perturbations.

Our implementation is publicly available under github.com/circle-group/PoseDFlow.

2. Related Work

We now review the literature on priors of human pose and recent methods for inversion. Note that, while several works such as MotionVAE [39], HuMoR [54], PhaseMP [57], NRMF [74] or [29, 35, 56, 59, 72], model human motion unconditionally as well as conditionally [17, 28], our scope is human pose. We present an extended review in our supplementary material.

Unconditional human pose priors. Human bodies have been modeled by unconditional priors of various different kinds including Gaussian Processes [70], VAEs [53], normalizing flows [20, 55], neural distance fields (NDFs) [15, 26, 60] and diffusion models [16, 32, 45, 46]. To the best of our knowledge, there is no unconditional flow matching model of human pose, let alone the geometric variant.

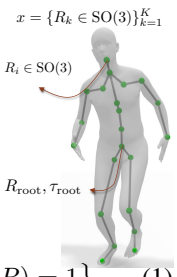
Riemannian flow matching (RFM). The flexible and general toolkit offered by RFM has been leveraged across numerous disciplines: in category level pose estimation by RFMPose [52], molecule generation by FoldFlow [9, 30], protein backbone generation by FrameFlow [71], generating materials by FlowMM [48], protein-ligand docking by FlowDock [50] & MATCHA [22], grasp pose generation by Equigraspflow [37], modeling brain connectivity by BrainFlow [9, 77], and for modeling statistical manifolds [9, 77], metal-organic structure prediction [33] and graph generation [12].

Guided flow matching. The Gaussian probability paths of flow matching have enabled a range of *training-free* methods for controlled generation, allowing a pre-trained model to generate samples that satisfy some target constraints [21, 24, 38, 76]. In particular, FlowGrad [42] inserts learnable control variables u_t at each integration step of the ODE and backpropagates a guidance loss through the trajectory. D-Flow [4] views controlled generation as optimizing the initial noise of a frozen generative model to minimize some terminal cost. Extending D-Flow, [31, 68] optimize the source distribution rather than a single point. OC-Flow [65] guides flow models by formulating flow-based generation as an optimal control problem with convergence guarantees. While OC-Flow has been extended to $\mathrm{SO}(3)$, and TFG-Flow [38] presented an $\mathrm{SO}(3)$-invariant control, our work constitutes the first general training-free guidance for RFM. To this end, we inherit D-Flow, due to its simplicity and the possibility of extension to the geometric domain of human pose we consider here.

3. PoseRFM

We now introduce our *expressive* and *generalizable* Riemannian flow matching model of 3D human pose learned from real human poses. To construct a flow on human poses that connects the target distribution to a source, we must first choose a suitable parametrization of articulation.

Geometry of human poses. The pose of a 3D articulated body $x := \{R_i \in \mathrm{SO}(3)\}_{i=1}^{K}\}$ is composed of K joints, each defined as a rotation $R_i \in \mathrm{SO}(3)$:



Definition 1 (SO(3))**.** *Rotations are elements of the special orthogonal group:*

$$\mathrm{SO}(3) = \left\{R \in \mathbb{R}^{3\times 3} : R^\top R = I, \det(R) = 1\right\}. \quad (1)$$

The power manifold of rotations $\mathcal{M} := \mathrm{SO}(3)^K = \mathrm{SO}(3)\times\cdots\times\mathrm{SO}(3)$, parameterizes all articulated poses. To geometrize these poses, we leverage the distance $d_{\mathcal{M}}$, exponential map Exp, logarithmic map Log and the Riemannian gradient grad on this product space [26, 74]:

$$d_{\mathcal{M}}(x, x') = \|d(R_1, R_1'), d(R_2, R_2'), \ldots, d(R_K, R_K')\|_2$$
$$\mathrm{Log}_x = \left(\mathrm{Log}_{R_1}, \mathrm{Log}_{R_2}, \ldots, \mathrm{Log}_{R_K}\right) \quad (2)$$
$$\mathrm{Exp}_x = \left(\mathrm{Exp}_{R_1}, \mathrm{Exp}_{R_2}, \ldots, \mathrm{Exp}_{R_K}\right) \quad (3)$$
$$\mathrm{grad}_x f(x) = \left(\mathrm{grad}_{R_1} f(x), \ldots, \mathrm{grad}_{R_K} f(x)\right) \quad (4)$$

where $(R_k, R_k') \in x \in \mathrm{SO}(3)^K$, and their operands are inherited from SO(3) as explained in our supplementary material.

We use the differentiable SMPL body model [43], $M(\boldsymbol{\beta}, x) := M(\tau, \boldsymbol{\phi}, x, \boldsymbol{\beta})$, parametrized by $K = 21$ joint rotations x, root translation $\tau \in \mathbb{R}^3$ and orientation $\boldsymbol{\phi} \in \mathrm{SO}(3)$, as well as $\boldsymbol{\beta} \in \mathbb{R}^{16}$ shape parameters. The joint positions $J \in \mathbb{R}^{3\times 22}$ are obtained via forward kinematics, while the vertices of the body mesh $V \in \mathbb{R}^{3\times 6890}$ via M.

PoseRFM: RFM model of human pose. We learn the manifold of plausible human poses through Riemannian flow matching [14] on a large corpus of data $\{x_i\}$, which we now introduce following the necessary definitions:

Definition 2 (Riemannian Flow [14])**.** *A time-dependent **flow** is a one-parameter family of diffeomorphisms $\{\psi_t : \mathcal{M} \to \mathcal{M}\}_{t=0}^{1}$ defined by integrating instantaneous deformations represented by a time-dependent vector field $u_t \in \Gamma(\mathcal{T}\mathcal{M})$ on the tangent space (Riemannian flow-matching field). ψ_t is defined by solving the following Riemannian ordinary differential equation (ODE) on $\mathcal{M}$ over $t \in [0, 1]$:*

$$\frac{d}{dt}\psi_t(x) = u_t\left(\psi_t(x)\right), \quad \psi_0(x) = x. \quad (5)$$

We also denote the flow map at $t = 1$ by $\psi_1 : \mathcal{M} \to \mathcal{M} : \psi_1(x_0) = x(1)$, a smooth cost $\mathcal{L} : \mathcal{M} \to \mathbb{R}_+$, and the source-point objective $\mathcal{L}(x_0) = \mathcal{L}(\psi_1(x_0))$.

Definition 3 (Probability path)**.** *Let $\mathcal{P}(\mathcal{M})$ denote the space of probability distributions on $\mathcal{M}$. A probability path $p_t : [0, 1] \to \mathcal{P}(\mathcal{M})$ interpolates between two distributions $p_0, p_1 \in \mathcal{P}(\mathcal{M})$ indexed by $t \in [0, 1]$. p_t is said to be **generated** by ψ_t if it* pushes forward $p_0 := p(x_0)$ *to $p_1 := p(x_1)$ following u_t,* i.e. *$p_t = [\psi_t]_{\#}(p_0)$. We define a smooth probability path between data p_1 and a reference p_0 as*

$$p_t(x) = \int_{\mathcal{M}} p_t(x \mid x_1)\, p_1(x_1)\, \mathrm{d}V(x_1), \quad (6)$$

where $p_t(x \mid x_1)$ is a geodesic Gaussian kernel with smooth schedulers $\alpha_t, \sigma_t > 0$ with $\alpha_0 = 0$ (see suppl. material).

Remark 1 (Velocity field)**.** *p_t satisfies a continuity (Liouville) equation on the manifold: $\partial_t p_t + \mathrm{div}_g(p_t u_t) = 0$, where u_t is the **velocity field** transporting probability mass along the manifold, where div_g is the divergence on $\mathcal{M}$.*

We are now ready to define Riemannian-FM (RFM).

Definition 4 (RFM)**.** *Given a probability path p_t, subject to the boundary conditions $p_0 = p_{\text{source}}$ and $p_1 = p_{\text{target}}$, as well as an associated flow ψ_t, Riemannian flow matching learns a* continuous normalizing flow *by directly regressing u_t through a neural network $v_w(x, t)$ parametrized by w.*

Definition 5 (Riemannian Conditional FM)**.** *The vanilla RFM objective is intractable as we do not have access to the closed-form u_t generating p_t. Instead, we regress v_w against a tractable* conditional vector field *$u_t(x \mid x_1)$, generating a* conditional probability path *$p_t(x \mid x_1)$ which can recover the target unconditional path by marginalization:*

$$u_t(x) = \int_{\mathcal{M}} u_t(x \mid x_1)\frac{p_t(x \mid x_1)p(x_1)}{p_t(x)}\, \mathrm{dV}_{x_1}. \quad (7)$$

Definition 6 (Generating conditional vector field)**.** *RFM defines a vector field $u_t(x \mid x_1)$ that generates $p_t(x \mid x_1)$ through a distance d by enforcing $d(\psi_t(x \mid x_1), x_1) = \kappa(t) d(x, x_1)$. The minimal-norm conditional field is [14]:*

$$u_t(x \mid x_1) = \frac{d}{dt}\log\kappa(t)\frac{d(x, x_1)}{\|\nabla d(x, x_1)\|_g^2}\nabla d(x, x_1). \quad (8)$$

For the geodesic distance $d := d_g$ and $\kappa(t) := 1 - t$, $\|\nabla d_g\|_g = 1$ and $d_g \nabla d_g = \nabla \frac{1}{2} d_g^2 = -\mathrm{Log}_x(x_1)$, giving

$$u_t(x \mid x_1) = \frac{1}{1 - t}\mathrm{Log}_x(x_1) \quad (9)$$

Algorithm 1 : PoseRFM training

Given: base & target distributions: $p(x_0), p(x_1)$
Initialize: parameters w of network $v_w(x, t)$ randomly
while not converged **do**
  sample noise $x_0 \sim p(x_0)$, target $x_1 \sim p(x_1)$
  sample time $t \sim \mathcal{U}(0, 1)$
  $x_t \leftarrow \mathrm{Exp}_{x_0}(t\,\mathrm{Log}_{x_0}(x_1))$
  $\mathcal{L}(w) \leftarrow \|v_w(x, t) - u_t(x_t \mid x_1)\|_g^2$
  $w \leftarrow$ `optimizer_step`$(\mathcal{L}(w))$
end while
Return: v_t

Algorithm 2 : Riemannian D-Flow

Given: Pre-trained flow model $v_w \approx u_t$, loss $\mathcal{L}$
Initialize: $x_0^{(0)} \in \mathcal{M}$ randomly, $v_0 = 0, m_0 = 0 \in T_{x_0}\mathcal{M}$, hyperparameters $\alpha, \beta_1, \beta_2, \varepsilon$
for $i = 1, \ldots, N$ **do**
  $x_1^{(i)} \leftarrow \texttt{solve_ode}(x_0^{(i)},\ v_w)$
  $\nabla\mathcal{L} \leftarrow \nabla_{x_0^{(i)}}\mathcal{L}(x_1^{(i)})$
  $x_0^{(i+1)} \leftarrow \texttt{RAdam_step}(x_0^{(i)}, \nabla\mathcal{L})$
end for
Return: x_1^N

With this choice of time scheduling Chen & Lipman [14] then define an explicit Riemannian conditional FM (RCFM) objective for learning as:

$$\mathbb{E}_{t,p(x_1),p(x_0)} \left\| v_w(x_t,t) + d(x_0,x_1)\frac{\text{grad } d(x_t,x_1)}{\|\text{grad } d(x_t,x_1)\|_g^2} \right\|_g^2,$$

whose gradient is the same as that of RFM. We use Eq. (9) to train our PoseRFM, *i.e.*, to obtain the parameters w of v_w. Here, $t \in \mathcal{U}(0,1)$ and $d(\cdot,\cdot)$ is the geodesic distance.

Training algorithm. At each training step, we sample a $x_1 \sim p(x_1)$ from the target distribution and a $t \sim \mathcal{U}(0,1)$ randomly. Using the conditional probability path construction, a sample $x_t \sim p_t(\cdot \mid x_1)$ is obtained. For this pair (x_t, x_1), the target conditional tangent vector field $u_t(x_t \mid x_1) = \dot{x}_t$ is computed in closed form on $\mathcal{T}_x\mathcal{M}$. We also compute the network output $v_w(x,t)$ and evaluate the loss, which is backpropagated to update the network parameters w. The pseudo-code is shown in Alg. 1.

Sampling / generation. Once trained, PoseRFM can sample novel poses by integrating the ODE on the manifold from $t: 0 \to 1$ using a manifold-aware ODE step, *e.g.*, Riemannian-Euler: $x_{t+\eta} \approx \text{Exp}_{x_t}\left(\eta\Pi_{\mathcal{T}_{x_t}\mathcal{M}}(v_w(x_t,t))\right)$. Some generation results are shown in Fig. 3.

Implementation details. For all applications, we parameterize the vector field $v_w(x,t)$ as a time-conditional neural network using simple a MLP (4 hidden layers, with a hidden dim size of 512), and Swish activation function. The input is a pose $x_t \in (\mathcal{M} := \text{SO}(3)^{\text{K}})$ with dimension 189 ($21\times3\times3$) concatenated with a time variable $t \in [0,1]$, and returns a vector in the tangent space $\mathcal{T}_{x_t}SO^{\text{K}}$. We train our model for 50,000 steps using AdamW optimizer [44] with a learning rate of 1e-3, weight decay of 1e-4, an exponential weight moving average (EMA) of 0.99, and a batch size of 4096. We use ReduceLROnPlateau scheduler, reducing the learning rate by 0.5 whenever validation loss stops improving for 5 evaluation steps.

Algorithm 3 `RAdam_step`

Given: $x_k \in \mathcal{M}$, $m_k \in \mathcal{T}_{x_k}\mathcal{M}$,$v_k$, $\nabla_{x_k}\mathcal{L}$, $\alpha, \beta_1, \beta_2, \varepsilon$
$g_k \leftarrow \text{grad}_{x_k}\mathcal{L} := \Pi_{T_{x_k}\mathcal{M}}(\nabla_{x_k}\mathcal{L}) \in T_{x_k}\mathcal{M}$
$m_k \leftarrow \text{PT}_{x_{k-1}\to x_k}(m_k)$
$m_{k+1} \leftarrow \beta_1 m_k + (1-\beta_1)\, g_k$
$v_{k+1} \leftarrow \beta_2 v_k + (1-\beta_2)\,\langle g_k, g_k\rangle_{x_k}$
$\hat{m}_k \leftarrow m_{k+1}/(1-\beta_1^t)$
$\hat{v}_k \leftarrow v_{k+1}/(1-\beta_2^t)$
$\Delta x_k \leftarrow -\eta\,\hat{m}_k/(\sqrt{\hat{v}_k} + \varepsilon)$
$x_{k+1} \leftarrow \text{Exp}_{x_k}(\Delta x_k)$

4. Riemannian D-Flow: Guiding PoseRFM

We now develop our algorithm, Riemannian D-Flow, to control the generation process of RFM models, in other words to invert our PoseRFM. We do so by extending D-Flow [4] to Riemannian manifolds following [14].

Given a pre-trained (frozen) PoseRFM model, $v_w(x,t) \approx u_t(x)$, represented as a neural network and some cost function $\mathcal{L}: \mathcal{M} \to \mathbb{R}_+$ and associated regularizers $\mathcal{R}$, our goal is to find likely samples $x \in \mathcal{M}$ that provide low cost $\mathcal{L}(x)$ and are likely under RFM's distribution $p(x_1)$. We formulate this as a *Riemannian source-point optimization problem*:

$$\min_{x_0\in\mathcal{M}} \big(\mathcal{L}(x(1)) := \mathcal{L}_{\text{data}}(x(1)) + \mathcal{R}(x_0,u)\big), \tag{10}$$

where $x(1)$ solves the Riemannian ODE in Eq. (5) with initial condition $x(0) = x_0$.

While our Riemannian D-Flow is a general framework, in this work, we make particular choices for objectives curated for human pose. While $\mathcal{L}_{\text{data}}$ is specified for each task distinctly in Sec. 5, we adopt a *trajectory regularizer* [66] common to all applications we consider:

$$\mathcal{R} := \mathcal{L}_{\text{traj}} = \sum_{i=1}^{K}\sum_{k=1}^{N}\big(3 - \text{tr}(x_{ik})\big), \tag{11}$$

where x_{ik} is the rotation of k^{th} joint at discrete timestep i. $\mathcal{R}$ explicitly forces the angle of rotations to be as small as possible, preventing wild trajectories and penalizing large, physically implausible rotations along the trajectory.

Algorithm and implementation details. Given a pre-trained flow, each iteration of Riemannian D-Flow solves the ODE forward in time via Euler integration, starting from $x_0 := x(0)$, and obtains $x_1 := x(1)$, where the loss $\mathcal{L}(x_1)$ is evaluated. The gradient $\nabla_{x_0}\mathcal{L}(\psi_1(x_0))$ is projected onto the tangent space of x_0 to obtain the Riemannian gradient. We then update the source point x_0 using Riemannian Adam [3]. This algorithm is summarized in Alg. 2. Other task-dependent hyperparameters are discussed in Sec. 5.

We now provide theoretical insights into the implicit biases of our Riemannian D-Flow.

4.1. Theoretical Analysis

We now provide a theoretical analysis that reveals an implicit regularization that comes from the choice of optimizing the cost as a function of the source point, where the gradient updates follow the data distribution $p_1 := p_1(x_1)$ by projecting the Riemannian gradient with the local data covariance matrix while respecting the curvature. Throughout, we assume the data distribution is supported away from the cut locus. All omitted proofs are provided in our supplementary material.

Theorem 1 (Tangent denoiser)**.** *For the data distribution $x_1 \sim p_1$ on $\mathcal{M}$, define at any $x \in \mathcal{M}$ the tangent random variable $\xi_x := \mathrm{Log}_x(x_1) \in \mathcal{T}_x\mathcal{M}$. The tangent denoiser μ at time t given by:*

$$\mu(x) := \mu_{1|t}(x) = \mathbb{E}\left[\xi_x \mid x(t) = x\right], \tag{12}$$

is the unique minimizer of the Riemannian flow matching loss:

$$\mathcal{L}(v) := \mathbb{E}\left[\|\xi - v\|_{g_x}^2 \mid x_t = x\right]. \tag{13}$$

Theorem 2 (Covariant derivative & covariance)**.** *The covariant derivative $(\nabla\mu)_v(x) : \mathcal{T}_x\mathcal{M} \to \mathcal{T}_x\mathcal{M}$ of the tangent denoiser is given by:*

$$(\nabla\mu)_v(x) = A_x[C(x)v] + R_x[v], \tag{14}$$

where $A_x[v]$ is a linear operator, $C(x) := \mathrm{C}_{1|t}(x)$ is a self-adjoint, positive semidefinite linear map $\mathcal{T}_x\mathcal{M} \to \mathcal{T}_x\mathcal{M}$, representing the covariance under the conditional distribution, and $R_x[v]$ is a Riemannian remainder*:*

$$C(x) = \mathbb{E}\left[\left(\xi_x - \mu_{1|t}\right) \otimes \left(\xi_x - \mu_{1|t}\right) \mid x(t) = x\right] \tag{15}$$

$$R_x[v] = \mathbb{E}\left[\nabla_v^{(x)}\xi \mid x\right]. \tag{16}$$

As $t \to 1$, $C(x)$ approaches the local data covariance under variance scheduling of geodesic kernels: $\sigma \to 0$.

Remark 2 (Euclidean space as a special case)**.** *In a Euclidean space with Gaussian densities where $\xi = \log_x(x_1) = x_1 - x$, $\nabla_v^{(x)}\xi = \nabla_v(x_1 - x) = -v$, the remainder becomes:*

$$R_x[v] = \mathbb{E}[-v \mid x] = -v. \tag{17}$$

In the D-Flow derivation this constant $-v$ cancels exactly with the same constant appearing from the score derivative, leaving no leftover curvature term: $R_x = 0$ (disappearing after cancellation). For AGPP in [4], $A_x = \alpha_t/\sigma_t^2 I$.

Corollary 1. *The covariant derivative of the marginal velocity field satisfies ($\circ$ denotes composition):*

$$\nabla u_t(x) = \frac{1}{1-t}\left(C_{1|t}(x) \circ A_x + R_x\right). \tag{18}$$

This is the drift of the adjoint ODE, determining the Riemannian adjoint, as we explain next.

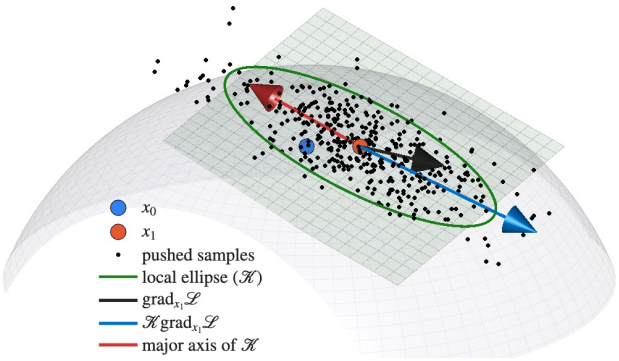


Figure 2. **Implicit bias:** The source point update steers the endpoint gradient towards the reachable subspace induced by the data.

Theorem 3 (Riemannian adjoint)**.** *Let $\Psi : \mathcal{M} \to \mathcal{M}$ denote the flow such that $x_1 = \Psi(x_0)$. Then the Riemannian gradients at the start and end points are related by the* Riemannian adjoint *(pullback map) $D_{x_0}\Psi(x_0)^*$:*

$$\mathrm{grad}_{x_0}\mathcal{L}(x_1) = D\Psi(x_0)^*\left[\mathrm{grad}_{x_1}\mathcal{L}(x_1)\right]. \tag{19}$$

Sketch of the proof. The result can be proven using the standard identities in Riemannian geometry, which we provide in our suppl. material for completeness. □

We finally prove the last bit characterizing the effect of the source-point update on the end-point.

Theorem 4 (Implicit bias in endpoint update)**.** *Consider a small optimization step updating the optimized source variable: $x_0 \longrightarrow \mathrm{Exp}_{x_0}\left(-\eta\,\mathrm{grad}_{x_0}\mathcal{L}(x_1)\right)$ where $\mathrm{grad}_{x_0}\mathcal{L}(x_1)$ is given in Eq. (19). As $\eta \to 0$ (infinitesimal change), the variation of the end-point $x_1 = \Psi(x_0)$, denoted δx_1 reads:*

$$\delta x(1) = -\eta \underbrace{\left(D_{x_0}\Psi(x_0)\, D_{x_0}\Psi(x_0)^*\right)}_{\mathcal{K}\text{ self-adjoint, PSD on }\mathcal{T}_{x_1}\mathcal{M}} \mathrm{grad}_{x(1)}\mathcal{L}(x(1)),$$

where $\mathcal{K} = D_{x_0}\Psi(x_0)\, D_{x_0}\Psi(x_0)^*$ resembles a local covariance on the endpoint gradient, explaining why the update is biased toward directions of high density, see Fig. 2.

We now discuss the importance of Thm. 2 and Thm. 4.

Remark 3 (Learning is **geometry-aware** and **data-adaptive**)**.** *Updating the source point during guidance leads to a* filtered update *on the end point, where the operator $\mathcal{K}$ is a projection onto the* reachable subspace *of the endpoint tangent space. When the flow generates the data distribution, this operator becomes the local covariance of the data manifold. Combined with the decomposition in Eq. (14) into a data-driven covariance term and an additional curvature-induced correction arising from the base-point dependence of the logarithmic map, suggests that the gradient updates of the terminal point are shaped by the* local data covariance *in the tangent space informed by the* curvature*. This is analogous to Euclidean D-Flow, yet mindful of geometry.*

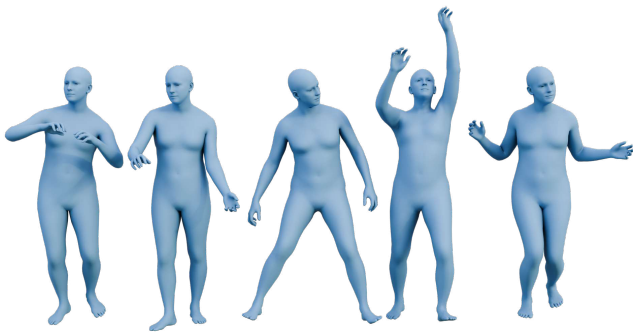


Figure 3. PoseRFM samples from unconditional generation.

5. Experiments

To evaluate its versatility, we evaluate PoseRFM on several inverse problems, covering both linear and non-linear pose problems: unconditional human pose generation, motion denoising from noisy joints, and human mesh recovery from 2D images (non-linear). We then conduct ablation studies to evaluate the contribution of each component in PoseRFM. Detailed hyperparameters, additional ablation studies and evaluations are available in our suppl. material.

Datasets. We use AMASS [47], a large motion capture database, to train our models using the same preprocessing and splits as prior work [26, 53, 54, 60]. We evaluate performance on the held-out AMASS test split, and further assess our method's generalization ability on HPS [25], EHF [53] and 3DPW [64], without additional finetuning.

Baselines. To ensure a fair comparison, we adopt the experimental setup from DPoser [46] for both training and evaluation. To study the impact of our geometric formulation in isolation, we also train a baseline Euclidean FM model, PoseFM, under the exact same setup as PoseRFM.

Metrics. For unconditional generation, we evaluate realism using FID (Fréchet Inception Distance) and d_{NN} (Nearest Neighbor distance) [26], and quantify diversity with APD (Average Pairwise Distance). Where a ground truth pose is available, we use MPJPE (Mean Per Joint Position Error) and MPVPE (Mean Per Vertex Position Error) to measure accuracy. We also report their Procrustes Aligned versions, PA-MPJPE and PA-MPVPE, which are rotation invariant and capture only body shape errors. Finally, PCK@50 (Percentage of Correct Keypoints) evaluates 2D joint localization accuracy, counting a joint as correct if its distance to the ground truth is within 50 mm.

5.1. Evaluations & Results

Pose generation. To generate realistic and diverse poses unconditionally, we randomly sample x_0 and use the midpoint ODE solver with 100 integration steps. Following [26], we sample 20 sets of 500 poses, and report the mean and 95% confidence intervals across runs in Tab. 1.

Our method achieves SotA performance in realism and fidelity, with the lowest FID and d_{NN}, indicating that the generated poses closely match real pose distributions. While not achieving the highest diversity, it remains competitive and outperforms several baselines. Methods achieving the highest APD (Pose-NDF [60] and NRDF [26]), do so at a substantial cost to realism, reflected by their higher FID and d_{NN} scores. Fig. 3 highlights the realism and diversity of our generated poses, suggesting a robust prior, capturing the distribution of plausible body configurations.

Table 1. Results for unconditional pose generation. Last three rows indicate our models.

Method	FID ↓	APD (cm) ↑	d_{NN} (rad) ↓
GMM [7]	$0.435^{\pm.017}$	$21.944^{\pm.102}$	$0.159^{\pm.001}$
VPoser [53]	$0.048^{\pm.002}$	$14.684^{\pm.138}$	$0.074^{\pm.000}$
GAN-S [19]	$0.201^{\pm.030}$	$10.914^{\pm.396}$	$0.098^{\pm.001}$
Pose-NDF [60]	$3.920^{\pm.034}$	$\mathbf{37.813}^{\pm.085}$	$0.838^{\pm.001}$
GFPose-A [16, 26]	$1.246^{\pm.005}$	$13.876^{\pm.116}$	–
GFPose-Q [16, 26]	$1.624^{\pm.002}$	$6.773^{\pm.112}$	$0.159^{\pm.000}$
FM-Dis [26]	$0.346^{\pm.007}$	$6.849^{\pm.199}$	$0.086^{\pm.001}$
NRDF [26]	$0.636^{\pm.007}$	$23.116^{\pm.105}$	$0.177^{\pm.001}$
Lie-HMR [32]	$0.825^{\pm.067}$	$\underline{23.999}^{\pm.602}$	–
DPoser [46]	$0.019^{\pm.001}$	$14.992^{\pm.123}$	$0.073^{\pm.001}$
PoseFM	$0.016^{\pm.001}$	$14.762^{\pm.176}$	$0.079^{\pm.000}$
PoseRFM (N=100)	$\underline{0.014}^{\pm.001}$	$15.481^{\pm.124}$	$\mathbf{0.069}^{\pm.001}$
PoseRFM (N=1000)	$\mathbf{0.013}^{\pm.001}$	$15.544^{\pm.167}$	$\underline{0.070}^{\pm.001}$

Pose completion. We now deploy our learned prior at reconstructing full human poses from partially observed or occluded poses, minimizing the total geodesic distance:

$$\mathcal{L}_{\mathrm{data}}(x) := \sum_{k\in\Omega} \cos^{-1}\left(\frac{\mathrm{tr}\left(x_k^\top x_k^{\mathrm{obs}}\right)-1}{2}\right) \qquad (20)$$

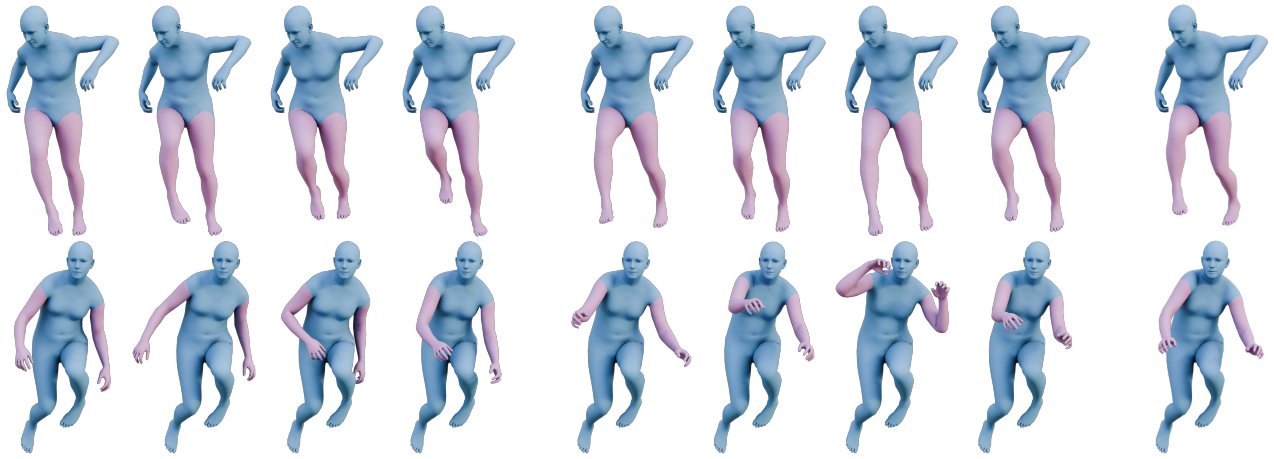
where x_k is the rotation of k^{th} joint of the generated pose. Ω is a known *mask* partially occluding the observation x^{obs}.

We evaluate this on the AMASS test split [47] with a subsampling rate of 10, and synthetically occluding four body parts: *left leg*, *legs*, *arms* and *torso*. For each occluded pose, we generate 10 hypotheses and evaluate their accuracy against the ground truth (GT) using the mean$^{\pm\mathrm{std}}$ of MPVPE across all hypotheses. To assess diversity among generated poses, we compute the APD over the 10 samples.

Our results in Tab. 2 shows competitive performance across all occlusion types, consistently balancing accuracy with diversity, with MPVPE being on par or better than DPoser [46], while also maintaining high diversity scores. Fig. 4 shows a qualitative comparison of completed poses, with occluded joints marked. Both DPoser [46] and PoseRFM produce realistic and diverse completions. This example also exposes a key limitation of the MPVPE metric, that assumes the GT pose as the only valid completion and penalizes other plausible alternatives. This motivates the need for a more suitable metric accounting for the distribution of plausible pose completions, akin to an FID.

Table 2. Pose Completion results under varying occlusion scenarios. We report MPVPE (mm) and APD (cm) metrics.

Method	Occ. left leg		Occ. legs		Occ. arms		Occ. torso	
	MPVPE ↓	APD ↑	MPVPE ↓	APD ↑	MPVPE ↓	APD ↑	MPVPE ↓	APD ↑
VPoser [53]	$200.51^{\pm 12.20}$	2.41	$221.34^{\pm 16.40}$	5.44	$206.72^{\pm 12.91}$	4.08	$58.71^{\pm 7.38}$	1.56
Pose-NDF [60]	$168.45^{\pm 8.66}$	1.95	$169.86^{\pm 6.12}$	1.97	$260.94^{\pm 4.81}$	1.20	$114.97^{\pm 5.43}$	0.93
CVPoser [46]	$128.04^{\pm 10.36}$	1.91	$134.35^{\pm 10.17}$	2.43	$162.82^{\pm 5.58}$	1.08	$51.23^{\pm 4.32}$	0.57
DPoser [46]	$\mathbf{78.31}^{\pm 27.13}$	6.53	$102.46^{\pm 25.39}$	7.75	$\mathbf{104.94}^{\pm 26.44}$	5.69	$44.60^{\pm 14.65}$	2.21
PoseFM	$102.60^{\pm 34.82}$	**8.77**	$129.04^{\pm 30.93}$	**9.94**	$140.88^{\pm 35.63}$	**7.85**	$51.86^{\pm 17.25}$	**2.70**
PoseRFM (ours)	$83.81^{\pm 29.07}$	6.02	$\mathbf{95.00}^{\pm 26.08}$	7.23	$107.36^{\pm 30.41}$	5.72	$\mathbf{39.75}^{\pm 9.12}$	1.21



(a) DPoser [46] (b) PoseRFM (ours) (c) Ground truth (GT)

Figure 4. Completed poses with legs (top row) and arms (bottom row) occluded. Visible and occluded joints.

Table 3. Motion denoising results. We report MPJPE (mm) under different Gaussian noise levels (σ= 4 / 10cm).

Method	AMASS [47]		HPS [25]	
	40 mm	100 mm	40 mm	100 mm
w/o prior	24.19	51.48	23.67	50.87
VPoser [53]	23.42	49.10	22.78	46.69
Pose-NDF [60]	22.13	46.10	21.60	47.50
MVAE [39]	26.80	–	–	–
HuMoR [54]	22.69	–	–	–
DPoser [46]	19.87	**33.18**	20.54	35.32
PoseRFM (ours)	**18.88**	34.68	**19.79**	**32.95**

Table 4. Human Mesh Recovery on EHF [53] dataset. We report PA-MPJPE (mm).

Method	from scratch	CLIFF [36] init.
w/o fitting	108.57	56.62
GMM [7]	58.32	51.02
VPoser [53]	58.08	49.39
GAN-S [19]	57.26	49.58
Pose-NDF [60]	57.87	49.50
NRDF [26]	57.38	49.27
DPoser [46]	56.05	49.05
PoseRFM (ours)	**54.85**	**47.24**

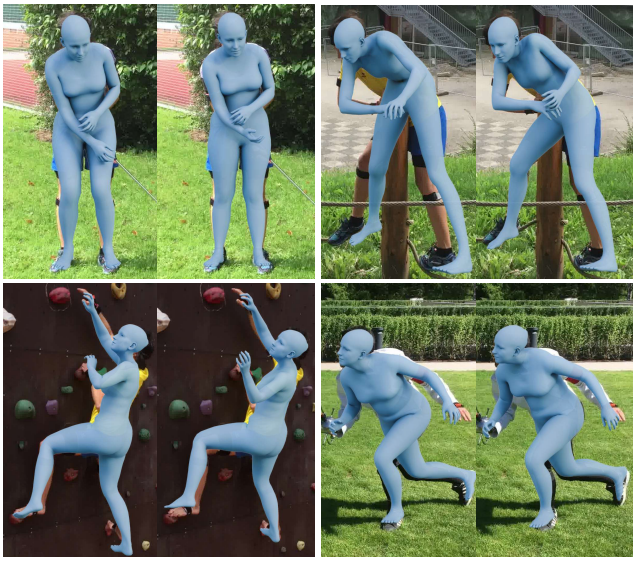
Motion denoising. Next, we consider motion denoising, which aims to recover temporally consistent human motion from noisy joint trajectories. We define the denoising loss:

$$\mathcal{L}_{\text{data}} := \mathcal{L}_{\text{joints}} + \lambda \mathcal{L}_{\text{smooth}} \tag{21}$$
$$= \|J_t - \mathcal{J}(M(\boldsymbol{\beta}, x_t^*))\|_2^2 + \lambda \sum_{k=1}^{K} \left(d_g\left(x_{t-1,k}, x_{t,k}\right)\right)$$

where J_t are some noisy joint inputs, x_t^* the estimated underlying clean pose, $\mathcal{J}$ extracts the joints from SMPL model, and $\mathcal{L}_{\text{smooth}}$ promotes temporal smoothness between consecutive poses.

We test this approach on the HumanEva subset from AMASS [47] and the HPS [25] dataset. All sequences are subsampled at 30 Hz and divided into 60-frame (2s) segments. We add Gaussian noise with $\sigma = 0.04, 0.1$ to the clean 3D joints and apply our conditional generation framework to recover the denoised poses. Our results in Tab. 3 show that PoseRFM consistently achieves superior performance over baselines across datasets and noise levels.

Inverse kinematics. Next, we test our framework on the non-linear human mesh recovery (HMR) task, which aims to estimate SMPL parameters $(\boldsymbol{\beta}^*, x^*)$ from 2D images and



DPoser [46] PoseRFM DPoser [46] PoseRFM

Figure 5. Results of HMR on in-the-wild images from 3DPW [64]. Fitting from scratch (top) and init. using CLIFF [36] (bottom).

observed 2D keypoints J_{obs}:

$$\mathcal{L}_{\text{data}}(x, \boldsymbol{\beta}) = \mathcal{L}_{\text{2D}} + \mathcal{L}_{\alpha} + \mathcal{L}_{\boldsymbol{\beta}} \quad \text{where :} \tag{22}$$

$$\mathcal{L}_{\text{2D}}(x, \boldsymbol{\beta}) = \text{diag}(\boldsymbol{\sigma})\rho(p_k^{\text{obs}} - \Pi(\mathcal{J}(M(\boldsymbol{\beta}, x^*)))) \tag{23}$$

$$\mathcal{L}_{\alpha} = \sum_{i \in (\text{elbow,knees})} \exp(R_i), \quad \mathcal{L}_{\boldsymbol{\beta}} = \|\boldsymbol{\beta}\|^2. \tag{24}$$

$\boldsymbol{\sigma}$ denote detection confidences, Π represents the perspective projection, and ρ the robust Geman-McClure function [7, 23]. If an estimated person segmentation mask is available, it is used to ignore spurious 2D joints.

We evaluate our method on the EHF [53] and 3DPW [64] datasets, by following the SMPLify [7] optimization pipeline. We begin with 2D keypoints predicted by ViTPose [69], initialize the pipeline with the mean pose, and optimize for camera parameters. We then optimize for the SMPL parameters $(\boldsymbol{\beta}, x)$ using Riemannian D-Flow. Additionally, we test an alternative initialization where the SMPL and camera parameters are first predicted by CLIFF [36] and used to initialize Riemannian D-Flow. We report the predicted joints accuracy using PA-MPJPE in Tab. 4. We advance the SotA under both initialization schemes, demonstrating the effectiveness of PoseD-Flow in solving non-linear inverse pose problems. The qualitative results in Fig. 5 highlight our method's ability to recover accurate poses in complex in-the-wild scenarios, often containing out-of-distribution and rare pose configurations. While PoseRFM generally fits better than DPoser [46], both methods occasionally struggle with unnatural hand and foot orientations. We provide an extended comparison with the current SotA in Tab. 5, including additional metrics for a more comprehensive evaluation.

Table 5. Detailed HMR evaluation with additional metrics.

Init.	Method	PA-MPJPE ↓	PA-MPVPE ↓	PCK@50 ↑
from scratch	DPoser [46]	56.05	53.67	61.60
	PoseRFM	**54.85**	**53.16**	**65.29**
CLIFF [36]	DPoser [46]	49.05	52.92	66.62
	PoseRFM	**47.24**	**48.67**	**71.40**

Table 6. Ablation study on the effect of geometric components.

Method	Geo. Loss	Traj. Loss	Occ. left leg		Occ. legs	
			MPVPE ↓	APD ↑	MPVPE ↓	APD ↑
PoseFM	✗	✗	102.60	8.77	129.04	9.94
	✓	✗	215.25	20.11	273.96	25.94
	✗	✓	141.66	5.92	148.77	6.29
	✓	✓	180.51	16.81	212.90	20.52
PoseFM + geometry	✗	✗	99.79	7.45	119.92	8.56
	✓	✗	91.00	6.34	115.58	7.45
	✗	✓	129.55	4.86	138.05	4.78
	✓	✓	90.46	5.79	115.67	6.53
PoseRFM	✗	✗	107.32	9.56	120.26	11.26
	✓	✗	91.80	8.07	107.28	10.02
	✗	✓	109.35	1.81	129.44	1.50
	✓	✓	83.81	6.02	95.00	7.23

Ablation study. Finally, we ablate on the contribution of each Riemannian component of our model on the pose completion task, including the model type (FM vs. RFM), the data loss (MSE vs. geodesic), and the trajectory loss. The results are summarized in Tab. 6. While the Euclidean baseline (PoseFM) is competitive, its accuracy is limited, reflected by its higher MPVPE. Adding geometric losses and regularization to this model further degrades performance for PoseFM. This is expected since PoseFM does not model the underlying manifold. In contrast, PoseRFM yields similar initial results but achieves significantly higher accuracy once geodesic and trajectory losses are incorporated, while still preserving output diversity.

6. Conclusion

We introduced PoseD-Flow, the first framework to unify Riemannian Flow Matching with training-free geometric guidance for human pose understanding. By learning an expressive pose prior (PoseRFM) directly on the pose manifold and developing a principled mechanism (Riemannian D-Flow) to invert it, we set a new state of the art across various challenging tasks, where robustness matters most. We theoretically show that the induced guidance dynamics inherit structure from data covariance and manifold curvature, yielding a natural inductive bias toward realistic poses. Empirically, PoseD-Flow sets a new state of the art across various challenging tasks.

Limitations & future work. Due to the backpropagation all the way to the source, our method is not real-time. Future work can take inspiration from FlowGrad [42] and OC-Flow [65] which provide ways to reduce memory usage. We plan to increase diversity by injecting stochasticity during inversion. Lastly, we will extend our work to motion [2, 74].

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